

VIMARISH K M

DATA ENGINEER

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SUMMARY

Data Engineer with a strong foundation in data architecture, pipeline development, and cloud-based data solutions, complemented by industry experience across data-intensive environments. Skilled in designing and implementing ETL workflows using Python, SQL, and cloud platforms (AWS, Azure, GCP). Proven ability to work with big data technologies such as Spark and Kafka, build data lakes and warehouses, and enable analytics through clean, reliable, and production-grade data infrastructure.

SKILLS

Data Technologies: Python, R, SAS, SQL, PL/SQL, PySpark, Pandas, NumPy, MySQL, MongoDB, Apache Hive, Apache Spark, Apache Kafka, Hadoop, Apache Airflow, SLURM, dbt, MLflow, DVC, CI/CD pipelines.

Tools: Databricks, Dockers, Github Actions, Git, Jupyter Notebooks, VS Code, JIRA, Pinecone.

Soft skills: Problem solving, Teamwork, Leadership, Initiative.

Cloud Technologies: AWS (S3, Redshift, Glue, EC2), Azure (Data Lake, Data Factory, Synapse), GCP (BigQuery, Cloud storage), Oracle, Web services (SOAP/REST).

PROFESSIONAL EXPERIENCE

Data Engineer, PricewaterhouseCoopers

Nov 2022 - Aug 2024

- Hands-on experience with Data bricks, Spark, and Python for processing large-scale structured and unstructured data across distributed environments.
- Built scalable, production-grade ETL pipelines using PySpark and AWS services to enable real-time and batch analytics.
- Designed and optimized cloud-native data lakes that support both historical reporting and advanced machine learning workflows.
- Proficient in data modeling and partitioning strategies to ensure high performance, cost-efficiency, and ease of querying in big data.
- Implemented robust data quality checks and CI/CD pipelines for automated testing, versioning, and deployment of data workflows.
- Implemented and maintained CI/CD pipelines using Jenkins and GitHub Actions, automating testing and deployment to accelerate development cycles.
- Proficient in Docker, cloud platforms (AWS S3, GCP, Azure), and integrating machine learning pipelines to support quantitative research.
- Worked closely with teams in Agile environments, to define business questions, requirements, timelines, and objectives, while proactively identifying opportunities for operational improvements.

Oracle Cloud Technical Consultant, PwC

Mar 2021 -Nov 2022

- Built OIC integration to streamline processes, eliminating the need for a manual program to improve workflow automation.
- Responsible for the development of multiple PL/SQL packages to stage MongoDB (JSON) data into Oracle ATP database, optimising data handling and processing.

PROJECTS

1. **AI Medical LLM** - (Python, Pinecone, OpenAI, Langchain, Groq, HuggingFace, Flask)-Developed an AI chat bot that answers medical queries by loading medical PDFs into the Pinecone vector database and utilising Groq for NLP. Deployed on AWS EC2 with GitHub Actions for CI/CD. [LLM Project](#)

2. **Pneumonia Detection - Lung X-rays** (Python, PyTorch, AWS S3, EC2, Streamlit, BentoML) - Built a deep learning pipeline to detect pneumonia from lung X-rays using VGG16-based CNN and custom CNN architecture. Automated S3-based image ingestion, applied real-time augmentation, and performed model evaluation on the best selected model. Deployed the application in S3 EC2 using git hub actions. [DLProject](#)

3. **Student Performance Prediction System** (Python, scikit-learn, Pandas, Flask) - Built an ML pipeline to predict student performance using multiple regression models. Implemented data ingestion, imputation-based transformation, hyper-parameter tuning, and model evaluation using RMSE/R². Deployed best model using Flask for real-time predictions. [MLProject](#)

EDUCATION

Master's in Data and Computational Science

University College Dublin, Ireland (Sep 2024 - Aug 2025)

Bachelor's in Electronics and Communication Engineering

PSG College of Technology, India (Aug 2017 - July 2021)

CERTIFICATIONS

- Certifications:** Data Engineer Associate, Oracle Cloud Platform Application Development 2021 Certified Specialist | 1Z0-1043-21, Oracle Cloud Platform Application Integration 2021 Certified Specialist | 1Z0-1042-21